2. Ice has latent heat of fusion $L = 3.34 \times 10^5\ \mathrm{J/kg}$. Meltwater at T_1 is not reused.

3. At pressure p and volume V, air has internal energy $U = 2.5pV$.

4. Good heat exchange keeps the air in the vessel isothermal during ejection.

5. Treat ejection as steps, each expelling volume u, with u far below the vessel volume. In each step, nozzle gas accelerates under the pressure difference and gains ΔE in kinetic energy and exits. Ignore its volume change while accelerating. Assume vessel pressure stays uniform during ejection; the outside pressure is atmospheric.

6. The electrical energy obtainable is 45% of the sum of ΔE.

7. If $|x| \ll 1$, then $\ln(1+x) \approx x$.

Using these assumptions, find the electrical energy obtainable from the iceberg.

VI. (15 points) Large vertical parallel metal plates A, B are d apart, at constant voltage U. A metal-coated table-tennis ball of mass m hangs between them. Nudge it towards one plate. It rebounds, hits the other plate, and repeats. Collisions are inelastic, with restitution coefficient e. Contact is very brief, during which electrostatic equilibrium is reached. The ball's charge q relates to plate voltage by $|q| = C_0 U$, where C_0 is constant. Assume the ball radius is far below plate spacing d, and its charge does not affect plate charge distribution. A long enough string makes the motion nearly horizontal and straight. Neglect initial kinetic energy at the first collision and air resistance. For the setup shown, find: